

CITY SCALE RETROFITTING ANALYSIS FOR AHMEDABAD



Mentors

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Github Repository:
[City Scale Retrofitting Analysis](#)

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Introduction & Project Motivation

Why is this project needed?

- Buildings account for nearly one-third of global energy consumption.
- Rapid urbanization has significantly increased building energy demand in Indian cities.
- Retrofitting existing buildings can improve energy efficiency and reduce carbon emissions.
- City-scale energy analysis helps identify high-impact retrofit opportunities for sustainable urban planning.

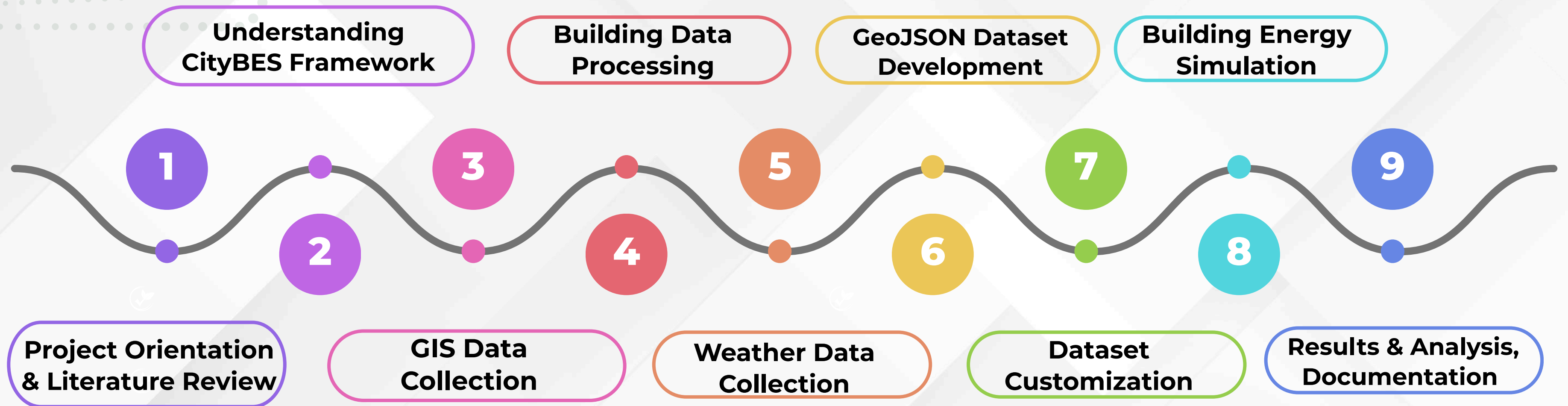
Why CityBES?

- CityBES enables Urban Building Energy Modeling (UBEM) using GIS and EnergyPlus.
- It allows energy simulation of thousands of buildings simultaneously.
- Supports evaluation of different retrofit strategies before implementation.

Objectives

- **Understand the fundamentals of Urban Building Energy Modeling (UBEM) and the CityBES platform.**
- **Prepare CityBES-compatible datasets for residential buildings in Ahmedabad.**
- **Collect, clean, and process GIS building data using QGIS.**
- **Develop GeoJSON datasets for building energy simulations.**
- **Integrate Ahmedabad-specific operational parameters using Python.**
- **Support city-scale energy analysis for future retrofit studies.**

Internship Workflow



Methodology

**Project Understanding &
CityBES input Requirements**

**Get the Energy Benchmark
values for the region**

**Building Data Collection
(Through QGIS using QuickOSM
for Ahmedabad) and Data
Availability Check**

**GeoJSON Preparation (Code)
(CityBES-Compatible Dataset)**

**GIS Data Processing
(Attribute Cleaning ,Height Estimation ,
Area Calculation , Finding lattitude &
longitude using formulas)**

**Python Dataset Customization
(Add additional_json Parameters
as per Indian Standards)**

**EPW Weather File Collection
(Ahmedabad TMYx) from Energy Plus
or can use ERA 5 Land Dataset and
extract using GEE**

**Upload to CityBES &
Baseline Energy Simulation**

What is CityBES?

- CityBES (**City Building Energy Saver**) is an open-source platform for Urban Building Energy Modeling (UBEM).
- It uses the EnergyPlus simulation engine to analyze building energy performance at the city scale.

What Does It Do?

- Simulates energy consumption of multiple buildings.
- Evaluates the impact of energy retrofit measures.
- Estimates EUI, heating, cooling, and electricity demand.

How Does It Help?

- Identifies energy-intensive buildings.
- Supports retrofit planning and energy-saving analysis.
- Assists in sustainable urban planning and decision-making.

Initial Approach: Retrofit Analysis Using CityBES

Objective

- Perform city-scale retrofit analysis for residential buildings in Ahmedabad
- Evaluate the impact of retrofit measures on building energy performance

Planned Retrofit Measures

- Roof insulation
- Wall insulation
- Window improvements
- HVAC efficiency upgrades
- Lighting improvements

Expected Outputs

- Building Energy Use Intensity (EUI)
- Cooling energy demand
- Heating energy demand
- Electricity consumption
- Retrofit energy savings

CityBES Data

Limitation for Indian Cities

While exploring CityBES, we observed:

- CityBES already provides datasets for several international cities.
- No pre-built datasets were available for Indian cities such as Ahmedabad.

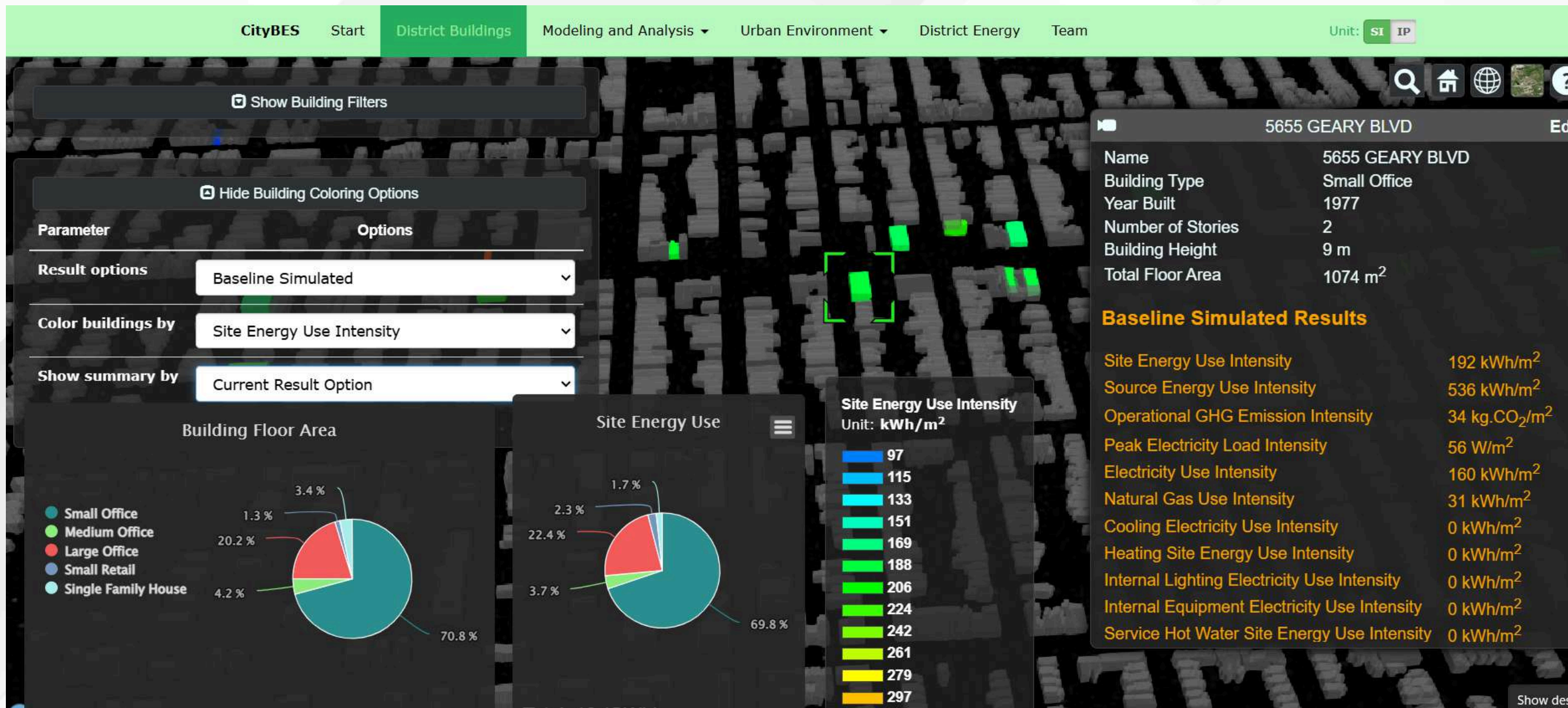
we decided to develop an Ahmedabad-compatible dataset by:

- Collecting GIS building data
- Processing building attributes
- Creating CityBES-compatible GeoJSON files
- Preparing EnergyPlus simulation inputs

Understanding CityBES framework and its Data Requirements

CityBES requires following inputs:

- Building footprints (geometry)
- Building height / number of storeys
- Building type
- Construction year
- Floor area
- GeoJSON format
- EnergyPlus weather (EPW) file
- additional_json file



Urban Building Data Collection

Data Sources Explored

- OpenStreetMap
- Google Earth Pro
- Ahmedabad 3D City Model
- VEDAS
- Bhuvan
- Copernicus Browser

Study Areas Initially explored:

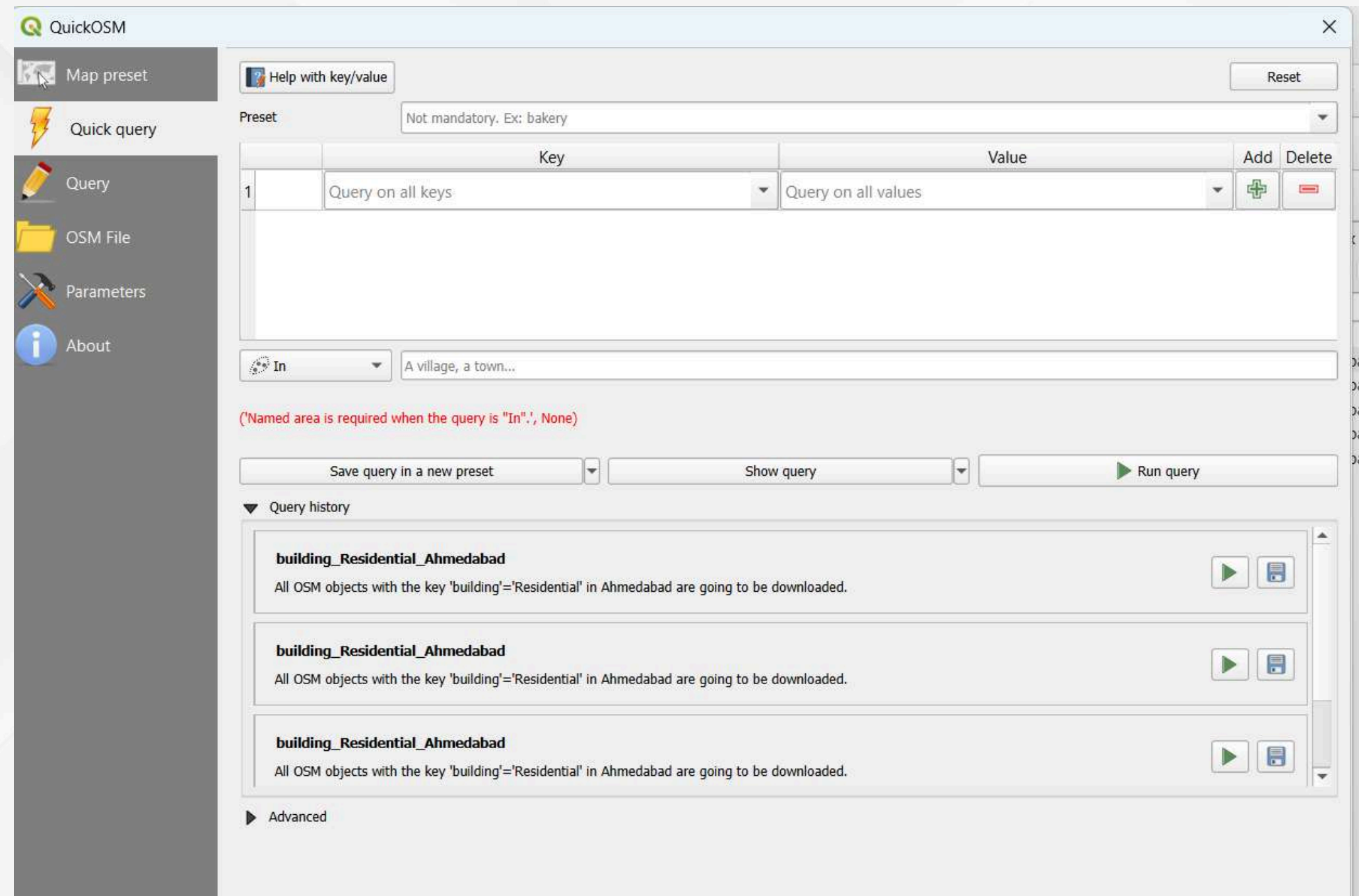
- Bopal
- Naroda
- Vastrapur

Final study areas:

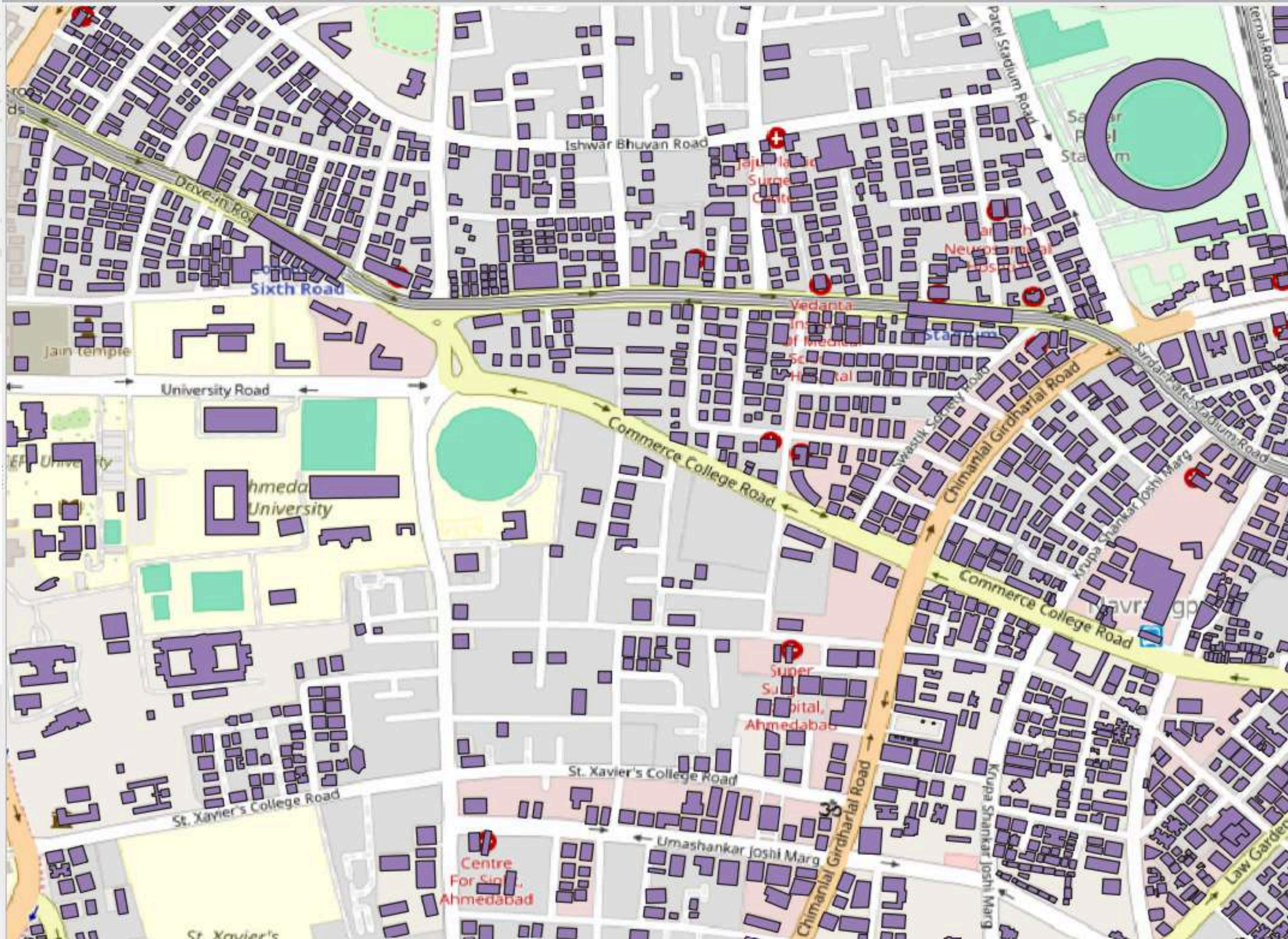
- Navrangpura
- Thaltej

Reason:

Availability of building:levels



QuickOSM plugin



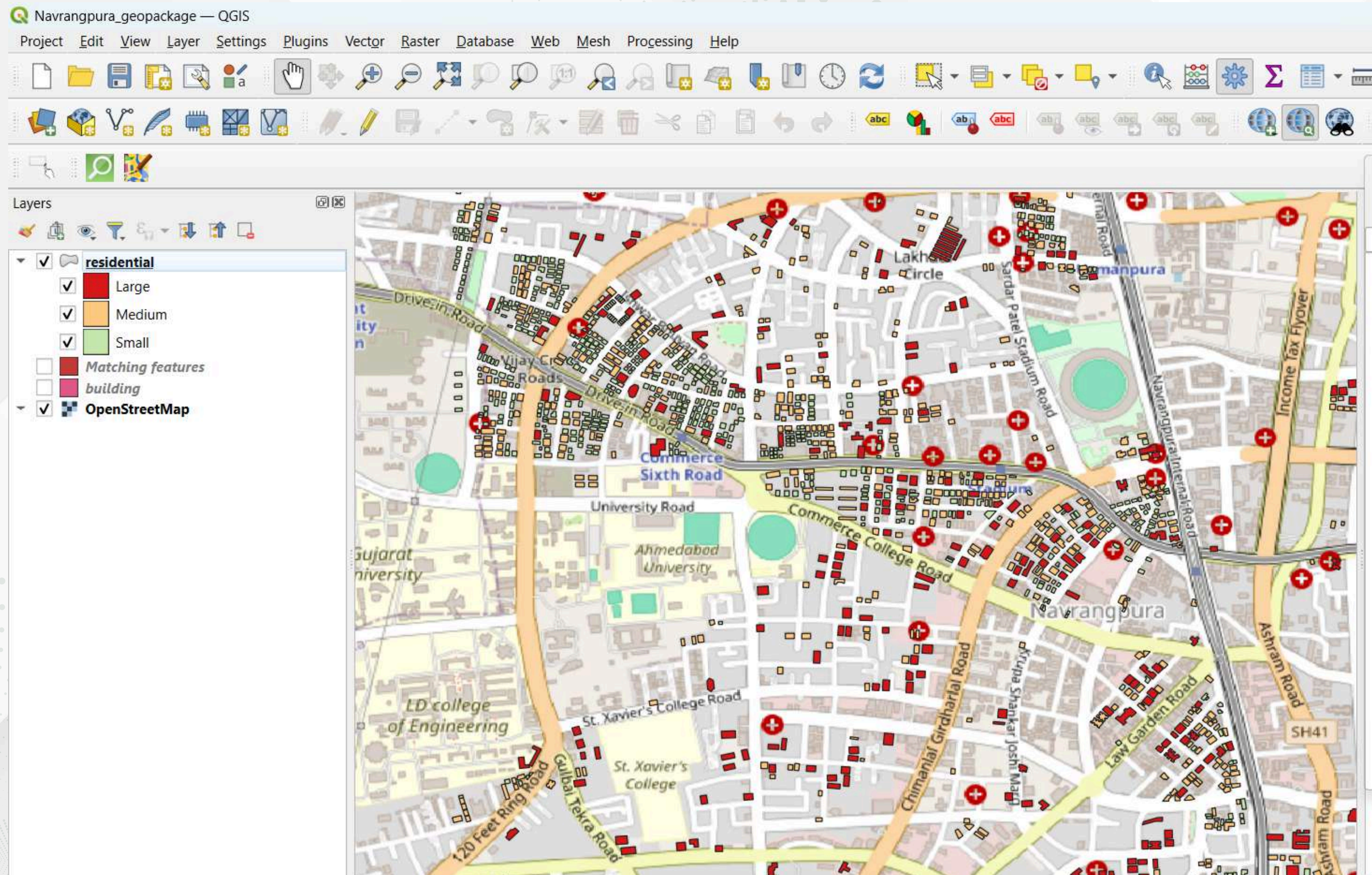
Extracted building foot printsx using QuickOSM plugin

GIS Data Processing Using QGIS

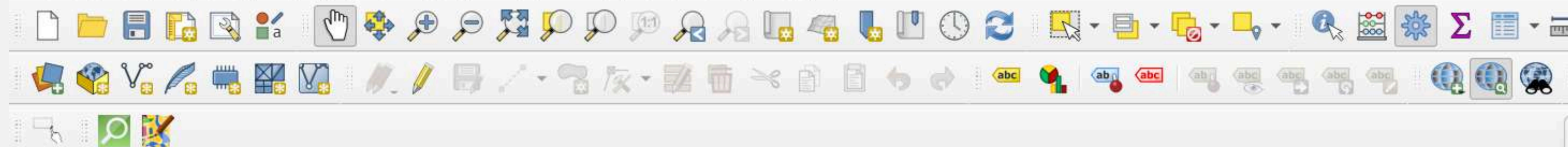
- Cleaning and filtering of the attribute table
- Calculation of building footprint area
- Estimation of building height using building levels (Height = Levels × 3 m)
- Computation of total floor area
- Standardization of building attributes (coordinates, construction year, etc.)
- Classification of buildings into Small, Medium, and Large categories for visualization and verification

	osm_id	building	building:levels ^	Area	longitude	latitude	Height	built_year	total_floor_area	size_class
1	487512912	residential	1	252.297	72.560422	23.038995	3	1990	252.297	Small
2	487512950	house	1	103.967	72.558871	23.040271	3	1989	103.967	Small
3	487513003	residential	1	438.776	72.561401	23.038643	3	2021	438.776	Small
4	524911395	residential	1	163.182	72.565033	23.033902	3	2004	163.182	Small
5	524912564	residential	1	45.561	72.565598	23.033448	3	1999	45.561	Small
6	529975298	residential	1	77.668	72.56283	23.036104	3	1990	77.668	Small
7	653186533	house	1	379.616	72.570091	23.028825	3	1992	379.616	Small
8	1292891373	residential	1	169.736	72.565473	23.038249	3	2000	169.736	Small
9	1292891384	residential	1	98.101	72.566244	23.039302	3	2006	98.101	Small
10	1293344394	residential	1	51.464	72.567353	23.033601	3	1997	51.464	Small
11	1293344409	residential	1	155.126	72.567716	23.033428	3	2018	155.126	Small
12	1298838300	residential	1	107.089	72.567706	23.033237	3	2004	107.089	Small
13	1302206683	residential	1	84.39	72.548107	23.044062	3	2007	84.39	Small
14	1302206684	residential	1	77.013	72.548279	23.043779	3	1989	77.013	Small
15	1302236931	residential	1	81.42	72.552635	23.041947	3	2006	81.42	Small
16	1302236985	residential	1	133.092	72.553175	23.042975	3	2012	133.092	Small
17	1302236996	residential	1	175.075	72.552238	23.042363	3	2017	175.075	Small
18	1302237016	residential	1	128.955	72.554654	23.041886	3	2009	128.955	Small
19	1302237035	residential	1	119.79	72.554468	23.041153	3	2017	119.79	Small
20	1302252365	residential	1	76.56	72.558366	23.040208	3	2023	76.56	Small
21	1302483187	house	1	83.849	72.567616	23.0303	3	2003	83.849	Small
22	1302483193	house	1	110.699	72.567948	23.030253	3	1995	110.699	Small
23	1302483198	house	1	354.286	72.568352	23.030712	3	2003	354.286	Small
24	1302483201	house	1	204.062	72.568	23.030376	3	2015	204.062	Small

Final attribute table after removing unnecessary columns



Classified buildings of Navrangpura with different colours

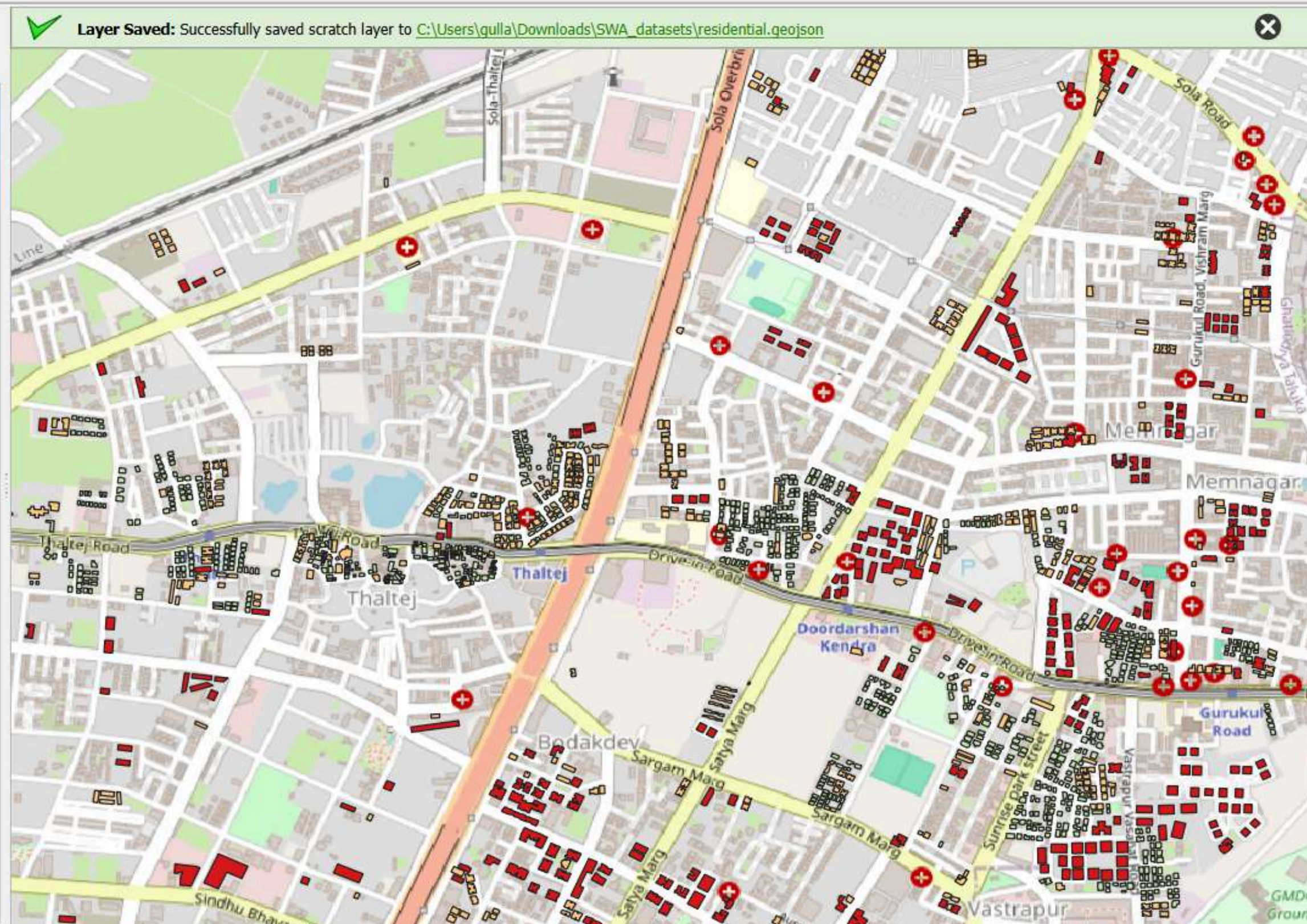


Layers



**'residential',
'house',
'apartments'**

- ☒ Large
- ☒ Medium
- ☒ Small
- ☐ building
- ☒ building
- ☒ OSM Standard



Classified buildings of Thatle with different colours

Collection of Ahmedabad Weather data (EPW)

Why is a Weather File Required?

EnergyPlus simulations require an EPW (EnergyPlus Weather) file containing hourly climatic data. The weather file directly influences the prediction of building heating, cooling, and overall energy consumption.

Weather File Selection Process

- Explored the official EnergyPlus Weather Data repository for Ahmedabad weather files.
- Since recent Typical Meteorological Year (TMY) datasets were not available, additional weather data sources were investigated.
- Evaluated weather files from Climate.OneBuilding.org and compared the available datasets.
- Selected the most recent and appropriate TMYx weather file for Ahmedabad for use in both CityBES and EnergyPlus simulations.

Collection of Ahmedabad Weather data (EPW)

Selected Weather File:

Weather_data_Ahmedabad.Intl.AP.426470_TMYx.2009–2023.epw

Source: [Climate.OneBuilding.org](https://climate.onebuilding.org)

Data Period: Typical Meteorological Year (TMYx) generated using weather observations from 2009–2023.

Why This File Was Selected

- Specifically represents Ahmedabad climatic conditions.
- Covers a recent 2009–2023 meteorological period.
- Compatible with both CityBES and EnergyPlus.
- Includes hourly data such as dry-bulb temperature, relative humidity, solar radiation, wind speed, and atmospheric pressure.

[illegible]

Ahmedabad-Specific Dataset Customization



Challenge Faced

- CityBES automatically assigns default building prototypes based on ASHRAE standards, which primarily represent North American buildings.
 - These default assumptions do not accurately reflect the operational characteristics of residential buildings in Ahmedabad, particularly thermostat settings and HVAC operation.
 - Therefore, the default building parameters needed to be customized before performing the energy simulations.
- 

Requirement of Additional_json file

- The GeoJSON file contains only building geometry and basic attributes, whereas CityBES requires additional operational parameters (HVAC, schedules, setpoints, etc.) for EnergyPlus simulations.
- If these parameters are not provided, CityBES assigns default values based on ASHRAE (American) standards, which are not suitable for Ahmedabad residential buildings.
- From the official CityBES documentation, it was understood that `hvac_system_type_id` represents predefined HVAC system types, not a unique ID for each building.
- Therefore, `hvac_system_type_id = 26` (Split System Air Conditioner for Single Zone) was assigned to all residential buildings as the most appropriate choice.
- A Python script was used to automatically append the `additional_json` field to every building in the GeoJSON dataset, making it simulation-ready for CityBES.



Additional Json code for customization of features

Code after integrating additional Json code

Upload to CityBES

CityBES Processing Workflow

- Upload the prepared GeoJSON dataset to the CityBES platform.
- Select the appropriate Ahmedabad EPW weather file.
- CityBES automatically converts the GeoJSON into an EnergyPlus Input Data File (IDF).
- The generated IDF is simulated using the EnergyPlus simulation engine.
- Simulation outputs are generated for subsequent energy analysis.

Challenges Faced

- **Missing building:levels information in several study areas.**
- **Identifying suitable study locations with sufficient building attributes.**
- **Preparing CityBES-compatible GeoJSON datasets.**
- **Understanding mandatory CityBES input parameters.**
- **Selecting the most appropriate Ahmedabad weather file.**
- **Ensuring compatibility between GIS datasets and CityBES simulation requirements.**
- **Debugging and validating Python scripts for dataset enhancement.**

Tools / Skills Used

Software & Tools

- QGIS
- CityBES
- EnergyPlus
- Python
- Visual Studio Code
- OpenStreetMap
- Google Earth Pro
- Microsoft Excel
- GitHub

Skills Developed

- GIS Analysis
- Spatial Data Processing
- GeoJSON Preparation
- Python Programming
- Data Cleaning
- Building Energy Modeling
- Technical Documentation
- Problem Solving
- Teamwork

Results

- **Successfully prepared CityBES-compatible datasets for Ahmedabad.**
- **Developed a structured workflow for Urban Building Energy Modeling.**
- **Processed and validated residential building datasets.**
- **Automated dataset customization using Python.**
- **Integrated Ahmedabad-specific operational parameters.**
- **Prepared simulation-ready GeoJSON files.**
- **Improved understanding of sustainable building energy analysis.**
- **Done baseline energy simulation for Navrangpura and Thaltej in Energy Plus**
- **Contributed to a project supporting future city-scale retrofit studies.**

Baseline Simulations

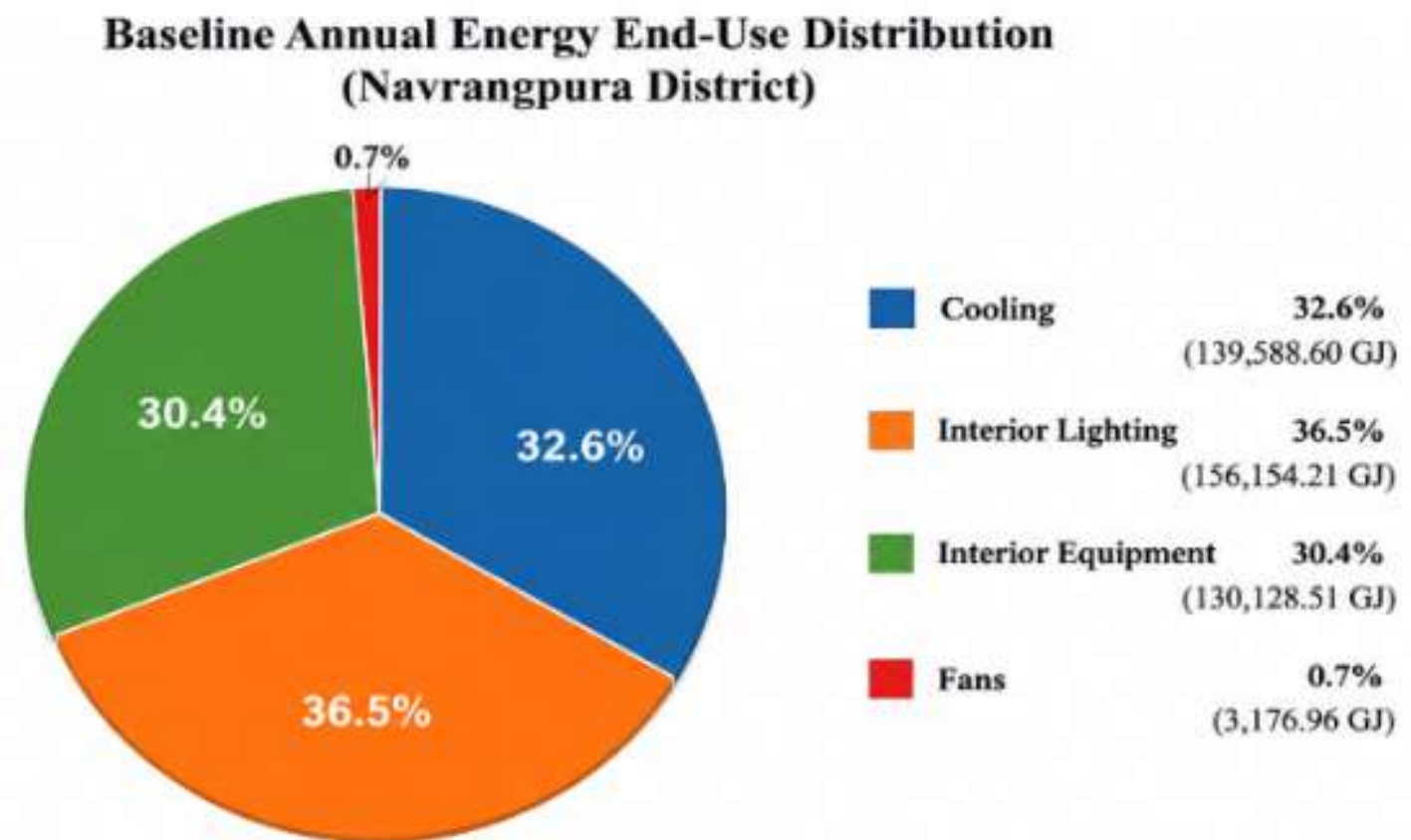
Parameter	Navrangpura	Thaltej
Total Building Area	825,269.59 m ²	1,191,137.13 m ²
Conditioned Area	412,634.79 m ²	595,568.57 m ²
Total Site Energy	429,048.28 GJ	651,649.56 GJ
Total Source Energy	1,358,795.89 GJ	2,063,774.14 GJ
Site Energy Intensity	519.89 MJ/m ²	547.08 MJ/m ²
Source Energy Intensity	1,646.49 MJ/m ²	1,732.61 MJ/m ²

Thaltej shows higher total site and source energy consumption than Navrangpura due to its larger building and conditioned area, while both regions exhibit similar energy intensity values, indicating comparable energy performance per unit area.

Major Energy Consumption

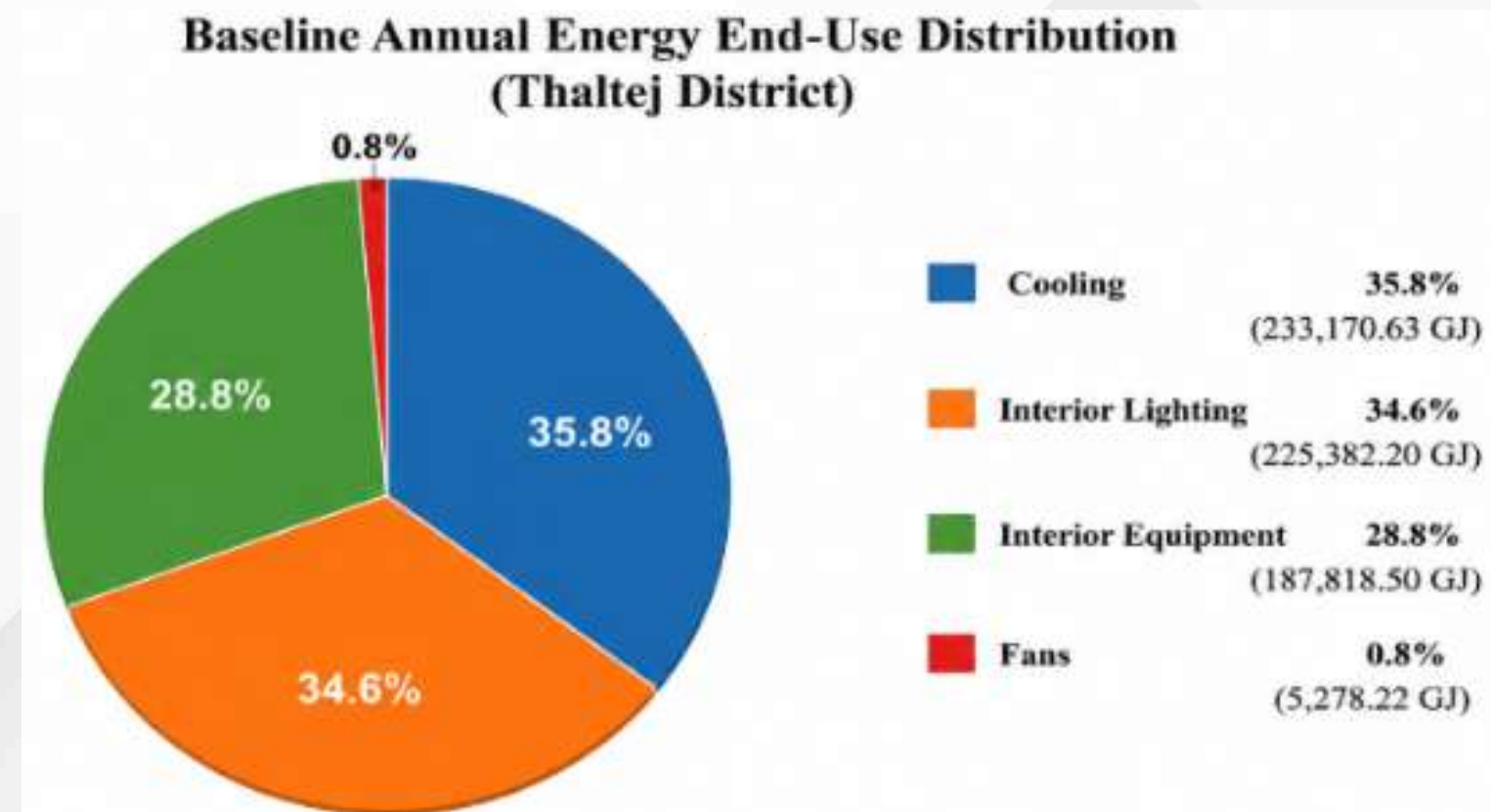
Navrangpura

- Interior Lighting – 156,154.21 GJ
- Cooling – 139,588.60 GJ
- Interior Equipment – 130,128.51 GJ
- Fans – 3,176.96 GJ
- Heating – 0 GJ



Thaltej

- Cooling – 233,170.63 GJ
- Interior Lighting – 225,382.20 GJ
- Interior Equipment – 187,818.50 GJ
- Fans – 5,278.22 GJ
- Heating – 0 GJ



Baseline Analysis

- The baseline EnergyPlus simulations indicate that cooling, interior lighting, and interior equipment are the dominant contributors to annual electricity consumption in both Navrangpura and Thaltej.
- Space heating demand is negligible due to Ahmedabad's hot climatic conditions, confirming that the city is predominantly cooling-dominated.
- Although Thaltej exhibits higher total energy consumption because of its larger building stock and floor area, the energy consumption patterns in both study areas are similar.
- The baseline simulation results identify cooling systems, lighting, and the building envelope as the primary targets for energy-efficient retrofit interventions.
- These baseline models established the reference case for evaluating the effectiveness of the proposed retrofit strategies.

7 Energy Conservation Measures (ECMs)

- Exterior Wall Improvement
- Roof Improvement
- Window Glazing Upgrade
- Solar Heat Gain Reduction
- Air Sealing Enhancement
- Cooling System Optimization
- Lighting Efficiency Improvement



Implemented ECM's

ECM	Baseline	Retrofit
Wall Assembly	200 mm Masonry Wall	60 mm PUF + 200 mm Masonry
Roof Assembly	150 mm RCC Roof	65 mm PUF + 150 mm RCC
Glazing Assembly	Single Glazing (U = 5.905 W/m ² ·K, SHGC = 0.75)	VIG (U = 0.60 W/m ² ·K, SHGC = 0.287)
Chiller	COP = 3.8	COP = 4.5
Lightling Loads	6 W/m ²	5 W/m ²

Retrofit Strategy

Why Polyurethane Foam (PUF)?

- Low thermal conductivity ($0.024 \text{ W/m}\cdot\text{K}$) for excellent insulation.
- High thermal resistance.
- Lightweight and easy to install.
- Low moisture absorption.
- Durable and low maintenance.

Why External Insulation?

- Reduces thermal bridges
- Preserves indoor space
- Easy retrofit installation
- Protects building envelope
- Lowers cooling energy demand

U-Value Calculations for Building Envelope

ISO 6946 – Thermal Resistance and Thermal Transmittance Calculation Method

For each layer in construction assembly,

$$\text{Thermal Resistance: } R = t / k$$

where:

- R = Thermal Resistance ($\text{m}^2\cdot\text{K}/\text{W}$)
- t = Material Thickness (m)
- k = Thermal Conductivity ($\text{W}/\text{m}\cdot\text{K}$)

$$\text{Total Thermal Resistance: } R_{\text{total}} = \sum R_{\text{layer}} + R_{\text{si}} + R_{\text{se}}$$

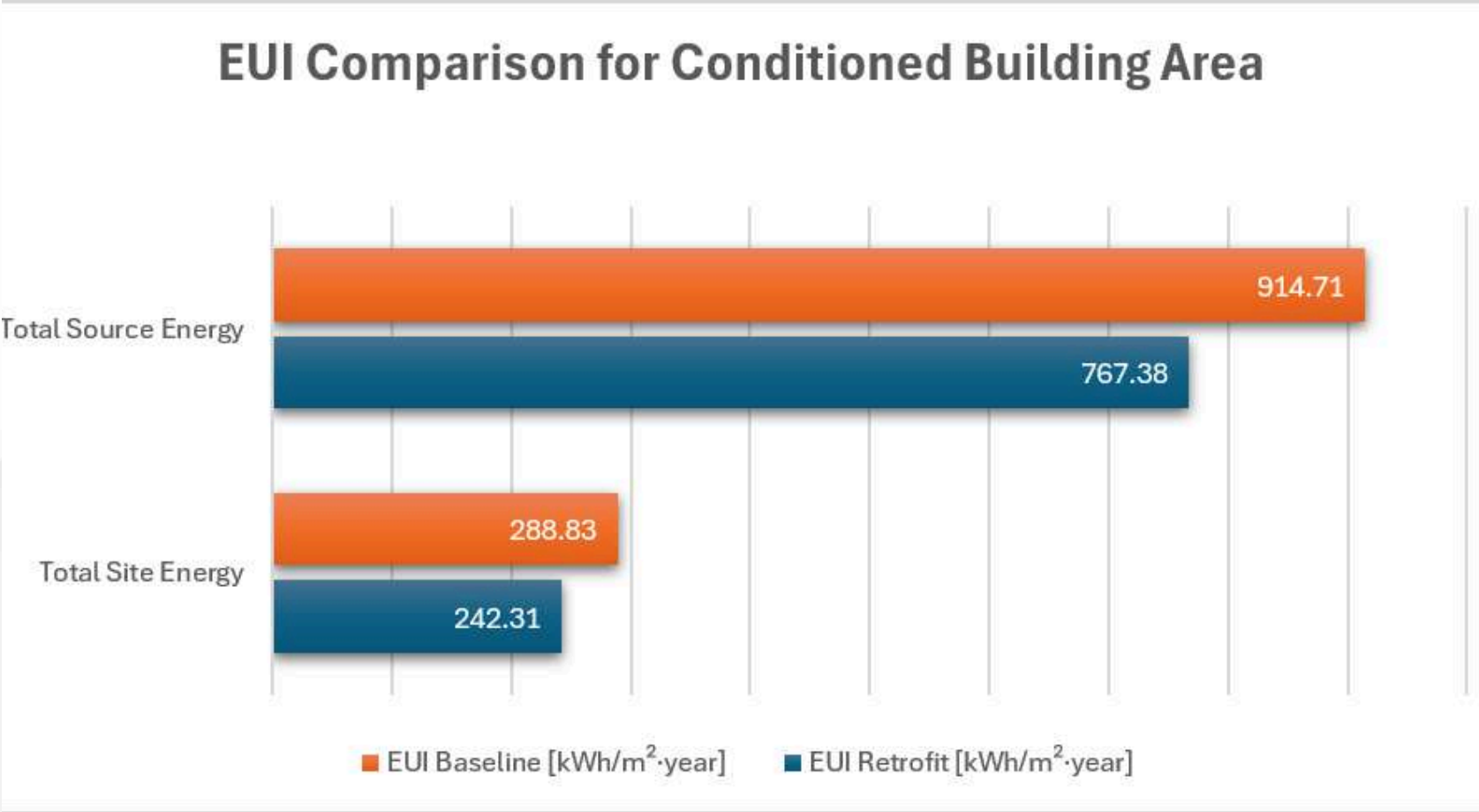
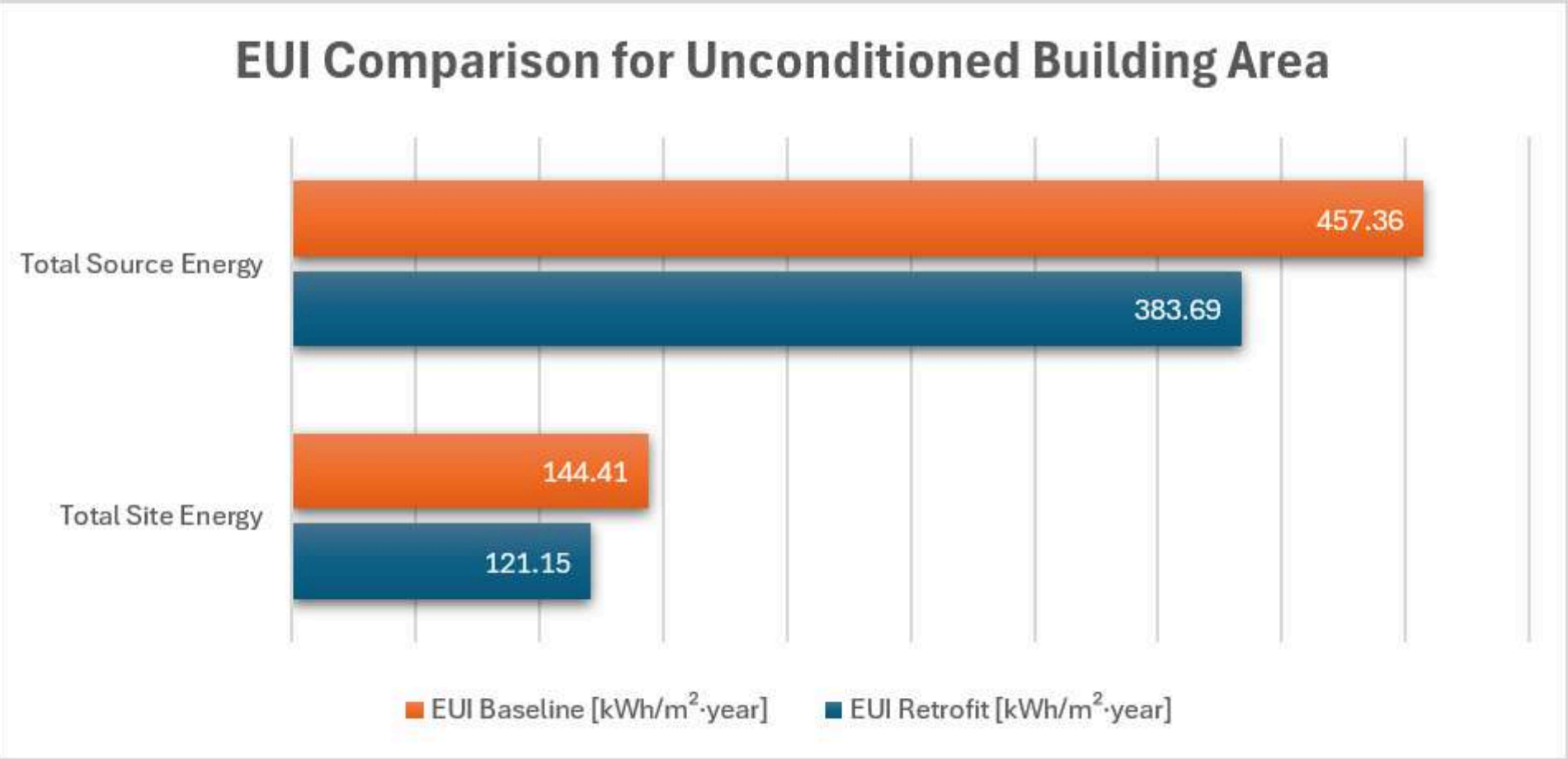
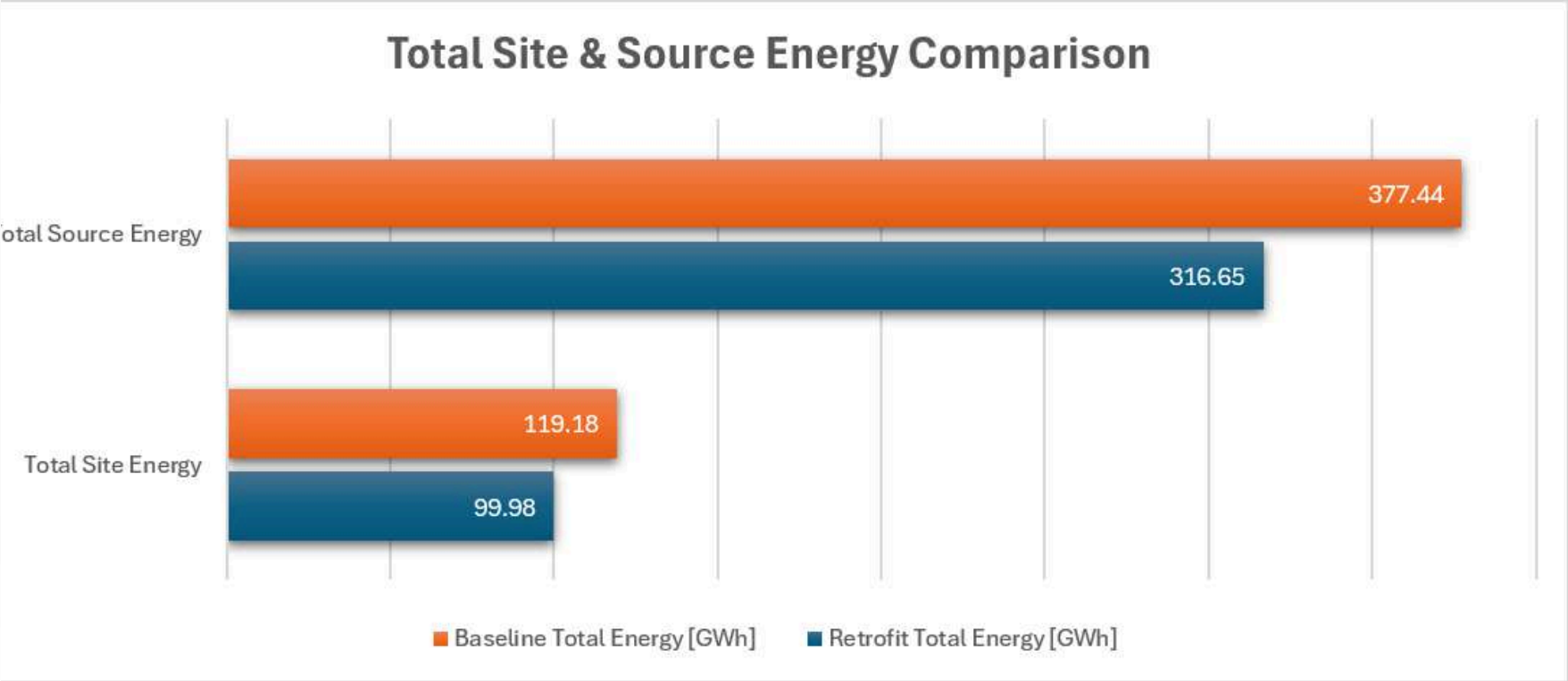
where:

- R_{si} = Internal air surface resistance
- R_{se} = External air surface resistance

$$\text{Overall Thermal Transmittance (U-value): } U = 1 / R_{\text{total}}$$

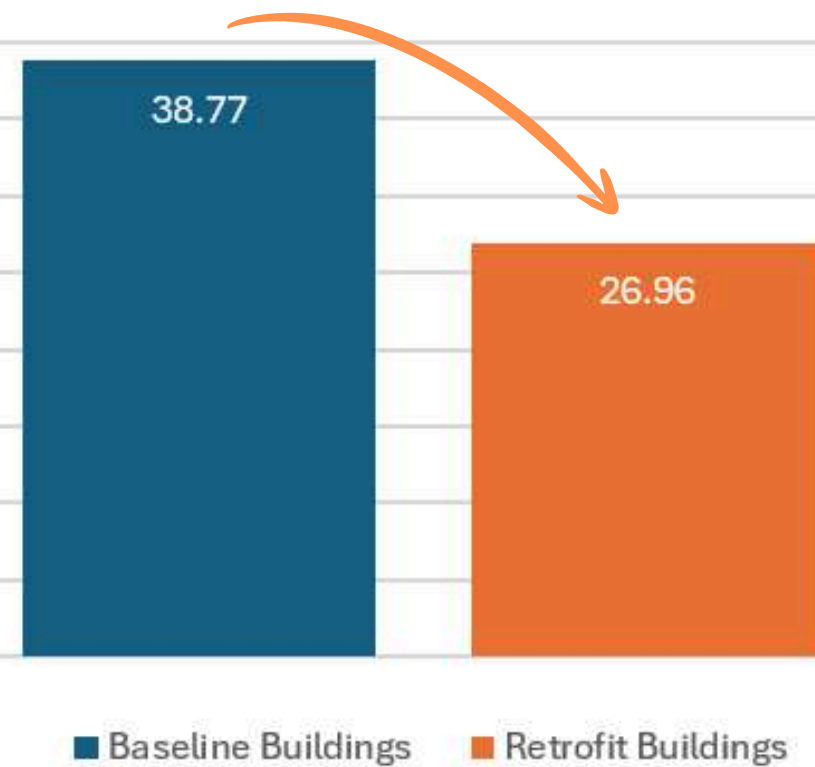
Result Comparisons (Navrangpura)

Category	Area [m²]
Total Building Area	825269.59
Net Conditioned Building Area	412634.79
Unconditioned Building Area	412634.79



Result Comparisons (Navrangpura)

Comparison of Baseline and Retrofit Cooling Load (Gwh)



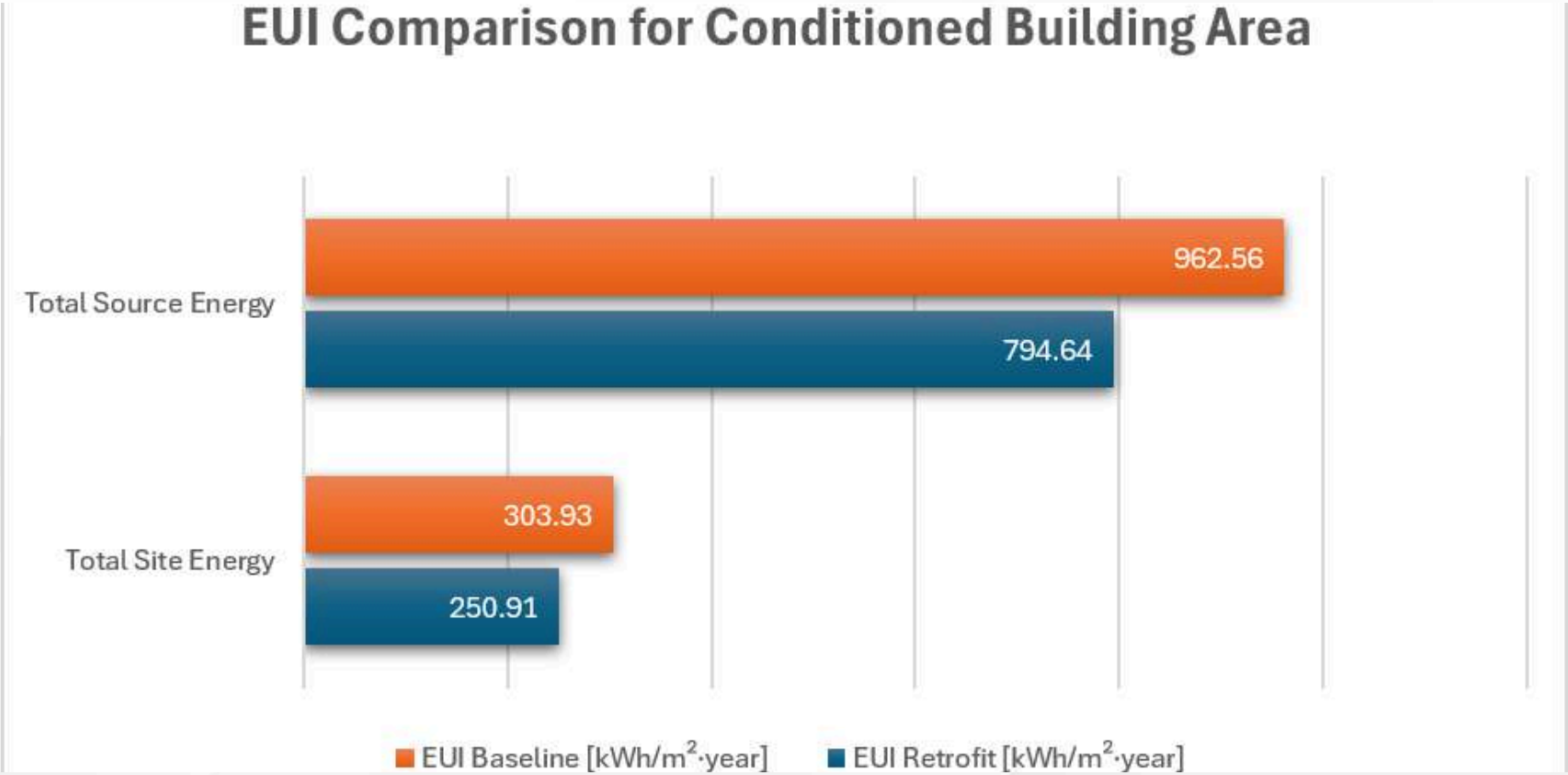
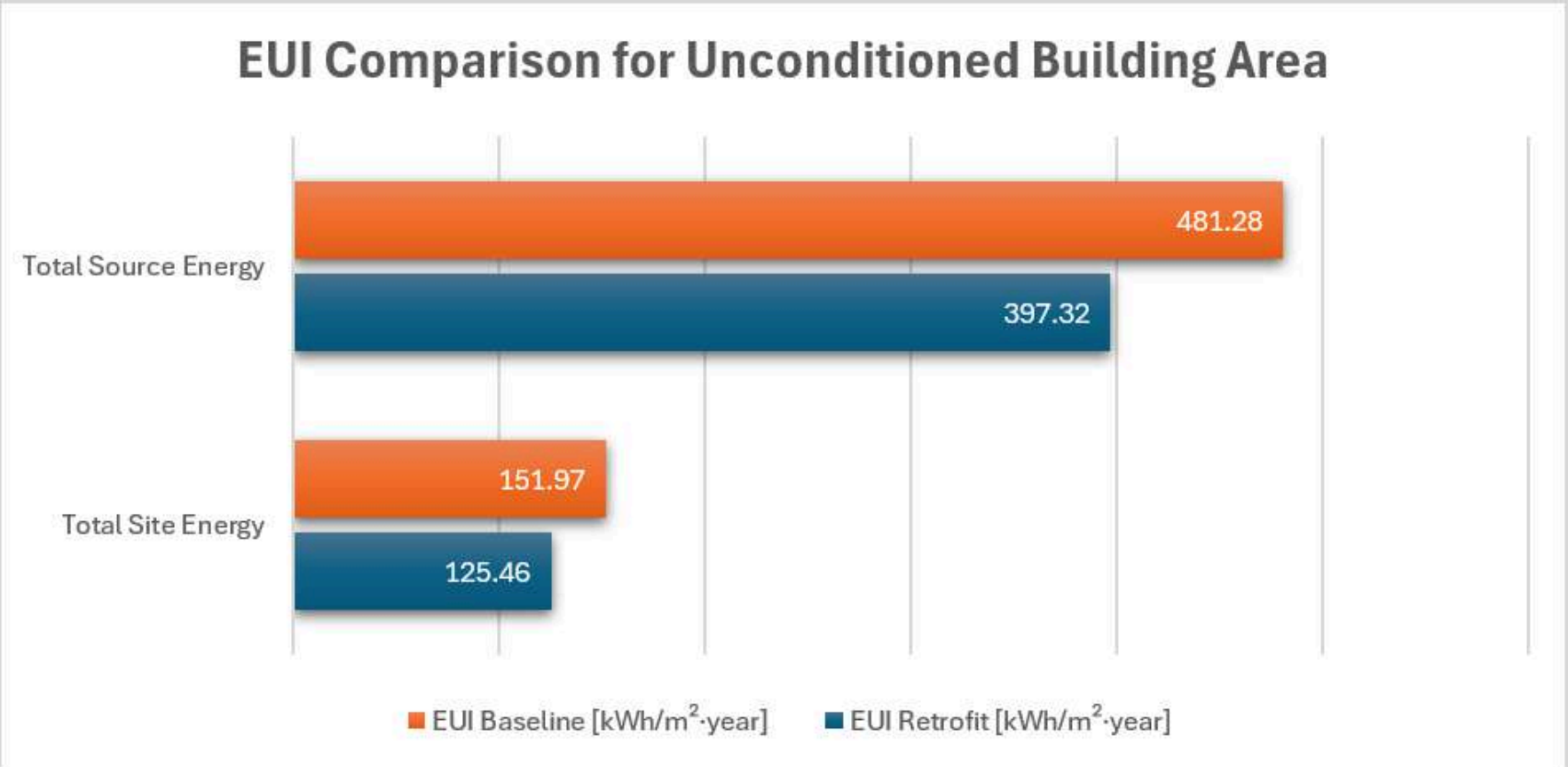
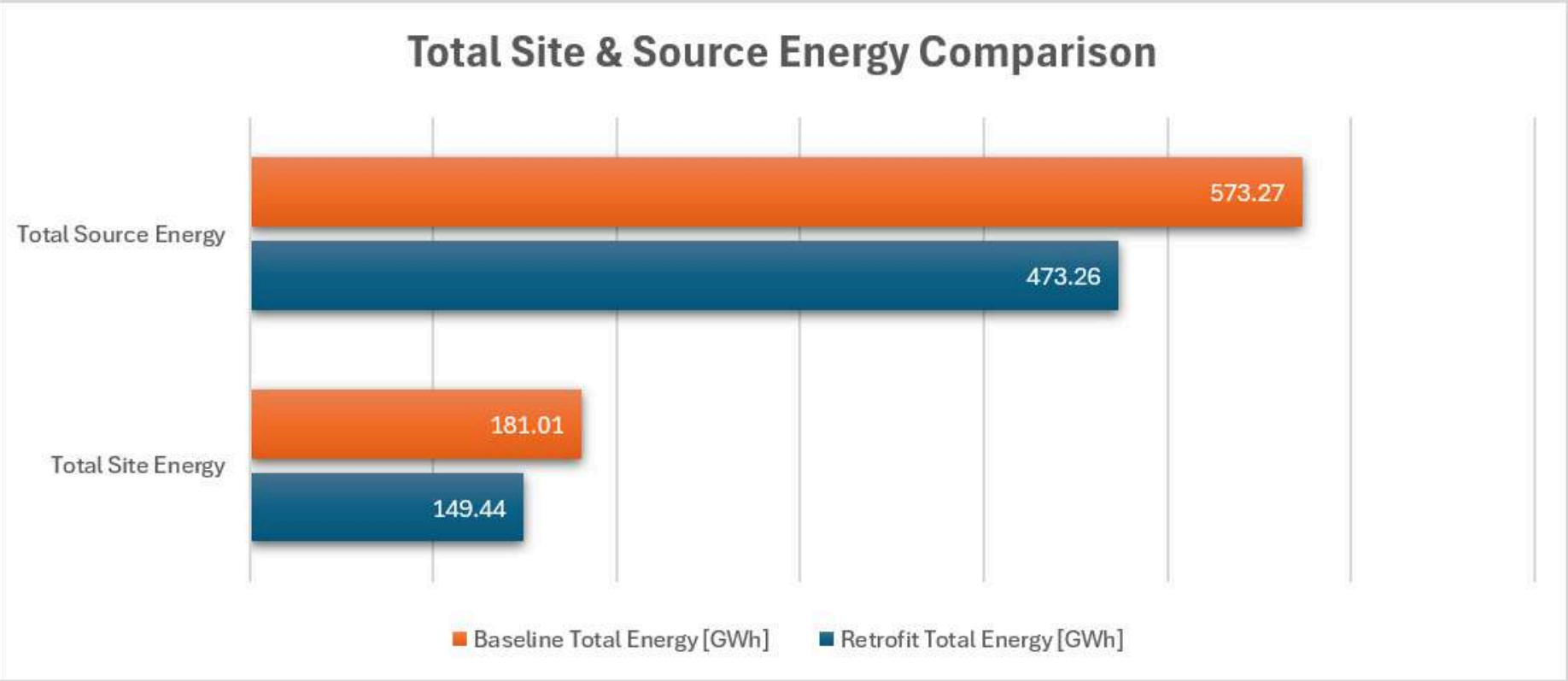
Comparison of Operation Carbon (tCO₂/year)



- Building envelope retrofitting significantly decreased the annual cooling electricity demand by **11.81 GWh (30.5%)**, demonstrating improved energy efficiency.
- Building envelope retrofitting lowered annual operational carbon emissions by **13,741.06 tCO₂/year (16.1%)**, demonstrating a significant reduction in the environmental impact of building operations.
- Retrofit reduced Total Site Energy by **16.1%** and Total Source Energy by **16.1%** compared to the baseline.

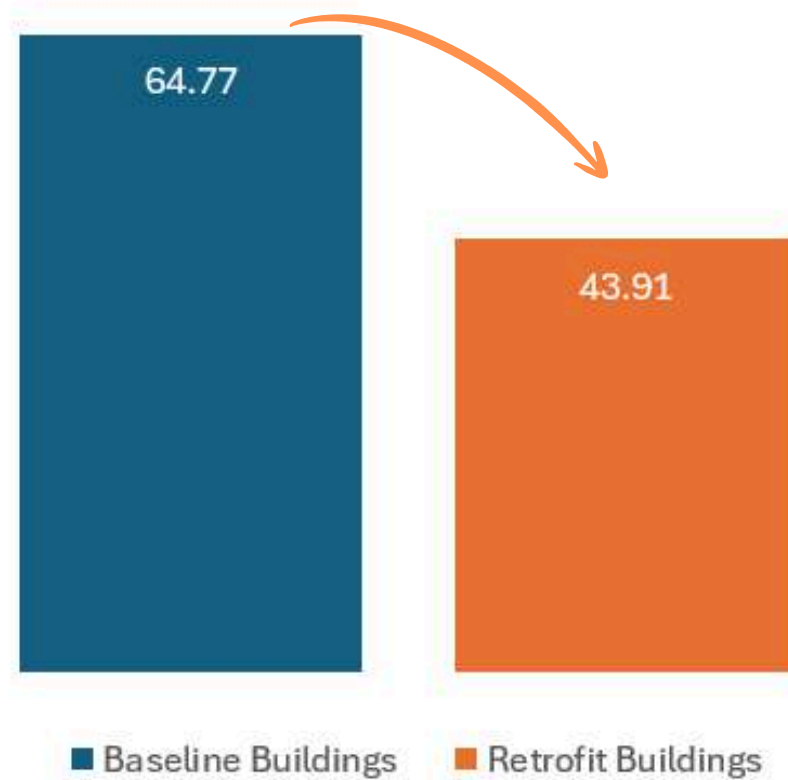
Result Comparison (Thaltej)

Category	Area [m²]
Total Building Area	1191137.13
Net Conditioned Building Area	595568.57
Unconditioned Building Area	595568.57

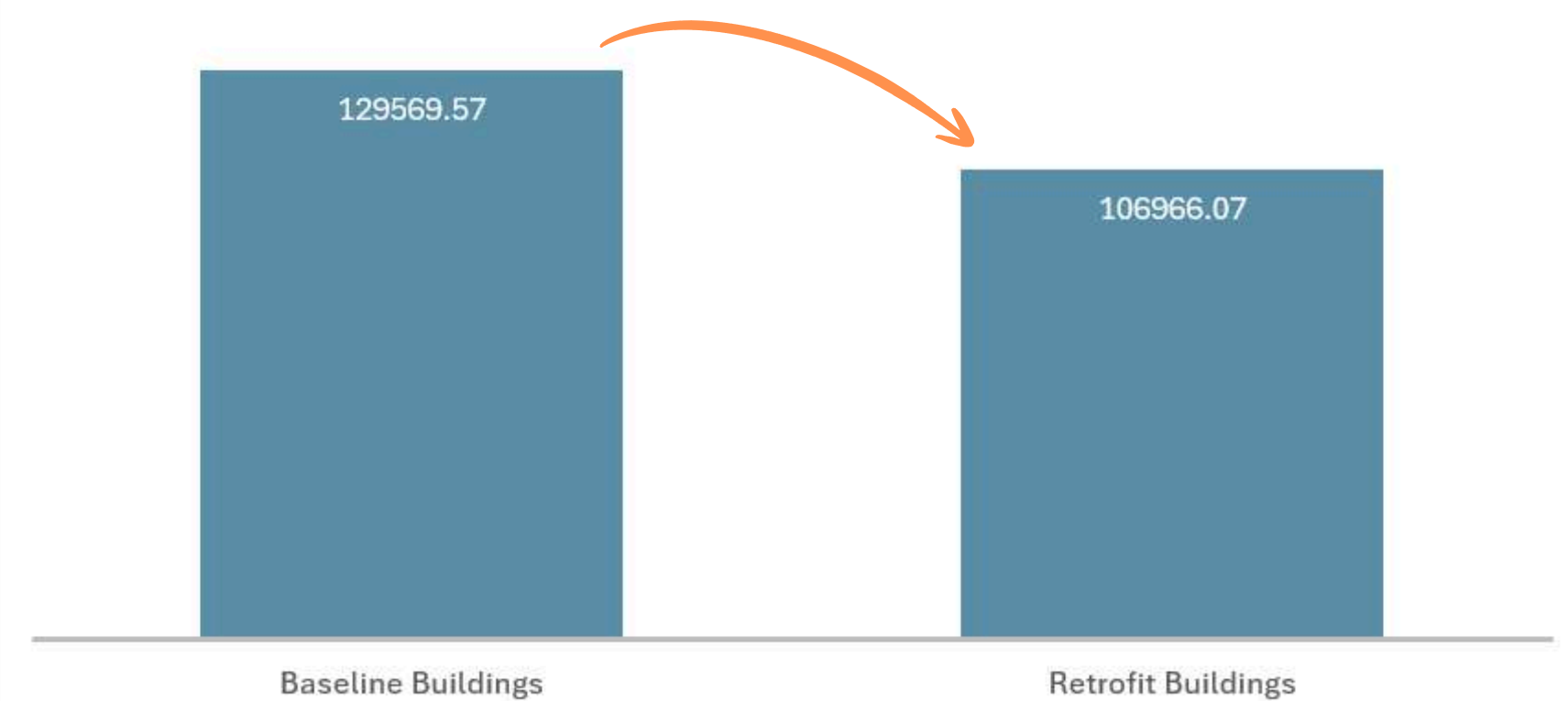


Result Comparison (Thaltej)

Comparison of Baseline and Retrofit Cooling Load (Gwh)



Comparison of Operation Carbon (tCO2/year)



- Building envelope retrofitting significantly reduced the annual cooling electricity demand by **20.86 GWh (32.2%)**, demonstrating improved energy efficiency.
- Building envelope retrofitting reduced annual operational carbon emissions by **22,603.50 tCO₂/year (17.45%)**, demonstrating a significant reduction in the environmental impact of building operations.
- Retrofit reduced Total Site Energy by **17.4%** and Total Source Energy by **17.4%** compared to the baseline.

Retrofit Performance Comparison - Navrangpura vs Thaltej

NAVRANGPURA			
Metric	Baseline	Retrofit	% Reduction
Total Site Energy (GJ)	429,048.28	359,939.78	16.10%
Total Source Energy (GJ)	1,358,795.89	1,139,929.29	16.10%
Site EUI (kWh/m²·yr)	144.41	121.15	16.10%
Source EUI (kWh/m²·yr)	457.36	383.69	16.10%
Cooling Demand (GWh)	38.77	26.96	30.50%
Operational Carbon (tCO₂/yr)	85,309.04	71,567.98	16.10%

THALTEJ			
Metric	Baseline	Retrofit	% Reduction
Total Site Energy (GJ)	651,649.56	537,968.87	17.40%
Total Source Energy (GJ)	2,063,774.14	1,703,747.42	17.40%
Site EUI (kWh/m²·yr)	151.97	125.46	17.40%
Source EUI (kWh/m²·yr)	481.28	397.32	17.40%
Cooling Demand (GWh)	64.77	43.91	32.20%
Operational Carbon (tCO₂/yr)	129,569.57	106,966.07	17.40%

Future Scope

- Extend the UBEM workflow to cover the entire Ahmedabad city and other Indian cities for large-scale energy assessment.
- Include additional building types, such as commercial, institutional, healthcare, and mixed-use buildings.
- Perform gap analysis by comparing simulated energy consumption with actual building electricity data to improve model accuracy.
- Develop an interactive GIS-based dashboard to visualize simulation results and support urban planning decisions.
- Conduct economic and environmental analyses, including life-cycle cost, payback period, and carbon emission reduction assessment.

THANK YOU